S(17): EXACT DEGREE AND RELATIVE FORMAL IMMERSIONS

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Theorem. For elliptic curves over number fields of exact degree17,

$$S(17)=\mathrm{Primes}(43).$$

The complete prime and support coverage is verified in final_coverage.json.
Failed exploratory calculations are not used as exclusions.

1. OCCURRENCE AND REDUCTION TO EXACT DEGREE

Mazur’s examples supply the primes through 7 after extension of scalars to any degree-17 field. For p=11,13,17,19 the genera of X_1(p) are 1,2,5,7. A rational cusp c and Riemann--Roch give a rational function with its only pole of exact order 17 at c, since 17>=2g-1. Hilbert irreducibility gives noncuspidal closed points of exact degree 17.

Explicit examples for p=23,29,31,37,41,43 are retained in occurrences.json, occurrence23.json and occurrences41_43.json. The three verification scripts check irreducibility of each field polynomial of degree 17, nonsingularity, and the exact prime order of (0,0) on the Tate curve

$$E=[1-c,-b,-b,0,0],$$

where

$$r=(x^2*y-x*y+y-1)/(x^2*y-x),\quad s=(x*y-y+1)/(x*y),$$
$$b=r*s*(r-1),\quad c=s*(r-1).$$

The examples at37,41,43 are from the local van Hoeij place table; the examples at23,29,31 are verified specializations. The equations and their verification, rather than completeness
of the place table, are what is used.

If p>43 belongs to S(17), the associated noncuspidal point of X_1(p) has degree dividing 17. Its degree cannot be 1 by Mazur. Hence its rational trace D is irreducible of degree exactly 17.

2. THE CORRECTED DYADIC BOUND

Use the previously established dyadic winding-quotient theorem, with its integral lattice statement 2M subset L subset M. Equality of the cotangent lattices is not assumed. Take m=34=2*17 and auxiliary modulus 181. For every u=1,...,180 the 34-by-180 matrix with entries

$$\mathrm{epsilon}(na)-\mathrm{epsilon}(nu/a),\quad 1<=n<=34,\ 1<=a<=180,$$

has rank 34 over F_2; epsilon indicates the upper half of nonzero residues modulo 181. prepare_bound.py retains the matrices, independently rebuilt and checked in verify_auxiliary.py. The intersection formula is valid for p>2*34*181=12308.

Thus the doubled-divisor, modulo-4 formal-neighborhood argument of the earlier dyadic theorem rules out a degree-17 trace supported at a single rational cusp of X_0(p), for p>12308. The doubling is necessary to retain validity with the possible factor-2 cotangent-lattice defect. Its proof uses that torsion in the kernel of reduction over Z_2 is killed by 2.

We improve the naive Hasse cutoff in degree 17. For p>66049, good reduction with residue degree <=16 is excluded by Hasse. If the residue degree is 17, an ordinary elliptic curve has a rational 2-torsion point, so its order is even. Hasse gives order <=131797, whereas an order divisible by such an odd p would be at least 2p>132098. For supersingular reduction, j=0 and the three twist orders over F_(2^17) are

$$130561=137*953,\ 131073=3*43691,\ 131585=5*26317.$$

bound_ss.m checks all twists and these orders. No prime >66049 divides

them. Thus no place above 2 can give good reduction for p>66049.

Additive reduction cannot support such prime torsion. If the torsion point lies in the identity component at a multiplicative place, then p divides 2^f-1 or 2^f+1. Such a place must have f=17, and is therefore the only place above 2, of ramification index 1. Consequently the trace on X_0(p) reduces entirely to one cusp. If there is no such place, the point maps nontrivially into the component group at every place, so again all conjugates reduce to the same cusp of X_0(p). After applying Fricke if necessary, this is 17*infinity. The dyadic argument excludes it. Since 12308<66049, we obtain p<=66049.

3. PUBLISHED GLOBAL EXCLUSIONS

Use Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1, Table 2, Proposition 4.1, Remark 4.4 and Corollary 7.5. Every prime considered is below 100000, within the proven range of their strange-character table.

The exact ledger applies Corollary 7.5, with its odd-degree hypotheses, and the appropriate Hecke/gonality criterion. For the ell=2 instances below 158, Remark 4.4 supplies the rational 2-torsion hypothesis. At 163 we instead use ell=3,a=3,n=7: the Kim--Sarnak gonality bound exceeds 7*17, and none of the possible noncuspidal orbit lengths divides 17. This avoids any rational 2-torsion assumption at 163.

The additional global candidates beyond the local levels below are 137,409,1021. At 137, Proposition 4.1 with ell=2,a=2,n=5 applies since 325*(137^2-1)/65536>85. Diamond 2 has order 34 modulo sign. An invariant irreducible divisor of degree 17 can therefore only consist of points with j=1728 (the exceptional orbit length is 17). cm_small.m rules out all such points of degree <=17: for E:y^2=x^3-x, Frobenius at 3 has trace0, and det(F^(4k)-1) is nonzero mod137 for every1<=k<=17.
A point over rational j=1728 is an Aut(E)-orbit of a nonzero p-torsion vector. A Frobenius cycle of length k<=17 on such an orbit would give a fixed vector of F^(4k), since Frobenius normalizes the order4 automorphism group. Thus these determinant checks exclude every possible residue-cycle length.

At 409 and 1021 the strangenesses are 12 and 30, respectively, and the subgroup in Corollary 7.5 has order 17. The asserted degree-17 point would give a rational noncuspidal point on the intermediate quotient, and hence on X_0(p). Mazur excludes rational p-isogenies for these p.

The remaining levels are
 47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127.
All gonality bounds subsequently used, except the elementary quotient arguments explicitly stated, are the unconditional Kim--Sarnak bound

$$\mathrm{gon}_{\mathbb{Q}}\, X_1(p) \geq 325(p^2-1)/65536.$$

No Selberg conjecture is assumed.

4. EXACT LOCAL CENSUS AND ORDINARY OBSTRUCTIONS

census_sources.m enumerates all j-invariants, twists, and p-torsion cyclic lines over F_(2^e), 1<=e<=17. When full rational p-torsion occurs, all p+1 lines are enumerated. Tate coordinates determine exact closed degrees and closed diamond orbits, eliminating duplicates exactly. Nonrational cusps have degree ord_(F_p^*/+-1)(2). All other cusp support is rational and has a unique rational cusp lift.

The canonical q-expansion and modular-unit data are the previously certified models (at127, the completed degree14 model export). Prior ordinary jets are reused only after unique exact matching with the new census. New sources use actual 3-isogenies and verified Hensel jets. At127 independent batches are combined only after every completion marker and all source metadata are checked.

The principal-divisor obstruction is exactly the one in the earlier proofs: evaluate the divisor alpha*(T_3-1-3*diamond3)R in characteristic 2. If this divisor is nonprincipal, the residue disk cannot contain D, since the operator kills rational Jacobian classes and all rational cusp contributions. The section spaces used are H^0(K^2) at47,53 and H^0(K) at the other levels. Multiplication tensors are verified using q-expansions beyond the weight-4 or weight-8 Sturm bound, as appropriate. Each full-rank matrix is retained and independently checked in Sage. A deficient rank is never treated as an exclusion.

For the ordinary operators, alpha is 1 at47,59,71; sigma^6-1 at61; sigma^12-1 at97, with (sigma-1)*Phi12(sigma) on the three ordinary closed orbits of size24 in degrees10 and14; (sigma-1)*(sigma^5+1) at101; and sigma-1 otherwise. The finite-image hypotheses follow from the rank-zero and strange-character certificates used in the earlier work.

At59 the first four jets at all29 rational cusps are116 independent vectors. Any degree17 cusp divisor has at most three cusps with multiplicity>=5. After a diamond translation and increasing high multiplicities if needed, all possibilities lie in the cases (17), (12,5),(11,6),(10,7),(9,8),(7,5,5),(6,6,5), together with the uniform four-jet block. The1625 positions of these cases give55335 nonzero relations, all of prefix cost>=58>17. verify_fi59.py independently recomputes every kernel and relation in Sage. This proves all cusp patterns without the much longer direct enumeration.

There are at most two ordinary support points. Total-degree-14 and 15 pair certificates are reused from degrees14 and15: their operator kills arbitrary rational cusp complements. New pairs exhaust totals 16 and17 and all relative diamond positions. Mixed operators at73 and 127 kill the supersingular or nonrational-cusp part, respectively. The coverage checker explicitly enumerates every support-degree multiset.

5. AN INTEGRAL RELATIVE FORMAL-IMMERSION LEMMA

Let S be smooth of relative dimension d over Z_2, and let L subset S be a smooth closed formal subspace of relative dimension r through a special-fiber point x. Let Phi:S->A be a morphism to a smooth group scheme, vanishing on L. If its differential has rank d-r at x and its kernel is the tangent space of L, then the completed zero fiber of Phi at x equals L.

Indeed choose formal coordinates u_1,...,u_r,v_1,...,v_(d-r) with L given by v=0. Choose d-r target coordinate functions with independent normal differentials. Since these functions vanish on L they are an invertible matrix over the complete local ring times v. Their common zero locus is therefore precisely L. This argument works integrally at2; it uses no p-adic integration denominator bound and no gonality inequality.

Here Phi is an actual integral Jacobian endomorphism composed with the Abel map. For a tame diamond subgroup <h> with quotient pi:X->Y, use

odd-prime torsion annihilators composed with h-1. Their image on J(Q) is zero because (h-1)J(Q) is finite. They also kill pi^*J_Y. Consequently Phi vanishes along each family F+pi^*Y^(r) when F is a fixed rational cuspidal divisor. More generally, a fixed local h-invariant divisor can be included: the remaining contribution is a rational cuspidal difference, killed by the odd-prime annihilator.

The exported cotangent matrices use the actual integral maps after Fricke transport. In the q-expansion basis these are, modulo2,
(D_h-1)*(T_q-D_q-q),
up to invertible diamond factors. Both support and cusp columns use the same map and basis. At a closed point the coefficient-coordinate matrix differs from the trace-pairing matrix by an invertible field basis change, so the ranks are the intrinsic cotangent ranks.

6. RELATIVE EXCLUSIONS AT53,61,73

At53 the only remaining supersingular divisor has degree13. The subgroup generated by diamond4 has order13 and acts freely on its geometric support. The divisor is a pullback of a rational point on Y=X_1(53)/<diamond4>. With any degree4 rational-cusp complement F, L=F+pi^*Y has dimension1. The maps for q=3,5 have transverse rank16 in all 23751 cusp configurations. Thus any rational D in the disk belongs to this pullback family and contains F, contradicting its irreducibility.

At61 the supersingular divisor has degree15. Diamond3 has order5 and acts freely, so it is a pullback of a degree3 divisor on the quotient. For every degree2 rational-cusp complement F, L=F+pi^*Y^(3) has relative dimension3. The transverse rank is14 in all465 configurations. Again D would contain F and cannot be irreducible.

At73 the difficult case is the supersingular degree6 divisor Z with a rational-cusp complement C of degree11. Put h=diamond2 of order9. Its order3 subgroup fixes the geometric points in Z. The stabilizer and nontrivial order3 tangent action give unique lifts of these points in the tame fixed locus over unramified extensions of Z_2. Their trace Z0 is an h-invariant local divisor of degree6. This is also checked by the rank6 transverse calculation at Z.

The 36 rational cusps form four h-orbits of size9. Write
C=R+pi^*E by extracting the minimum multiplicity on each orbit.
Then r=deg(E) is 0 or1, deg(R)=11-9r>0, and

L=Z0+R+pi^*Y^(r)

is a smooth local family of dimension r on which Phi vanishes. The four supersingular translates and every normalized cusp pattern are checked by relative_prefixes.cpp. There are760178194 patterns for each translate, independently counted by dynamic programming. The kernel dimension is exactly r: the only relations are the full h-fiber directions. The integral relative lemma forces D into L. Its fixed rational cusp contribution R is nonzero, a contradiction.

The other pure supersingular/nonrational-cusp cases at73 have support degrees12,9,15. The existing actual-Jacobian certificate shows diamond2 acts trivially on rational2-torsion. Thus the torsion class
y=[h(D-C)-(D-C)] reducing to0 is0, by its norm relation and odd order9. The gonality bound exceeds17+c for c=5,8,2. Hence hD=D. Its possible noncuspidal orbit lengths are9 or3, neither compatible with17.

7. RELATIVE EXCLUSION AT103

The cusp test is performed on Y=X_1(103)/H with [Delta:H]=17. Every normalized degree17 rational-cusp pattern passes except the sum of all17 rational cusps, each with multiplicity1. Independent enumeration of the relation space proves that this is the sole deficient pattern. It is exactly the pullback of the rational cusp on X_0(103).

The additional map (diamond5-1)*(T_3-3-diamond3), on the integral Jacobian of Y, kills J_Y(Q) since103 is non-strange, and kills the pullback of J_0(103). Its cotangent rank at this fiber is16. The relative lemma therefore forces any rational divisor in this disk to be a pullback of a rational point of X_0(103). Rationality follows also by uniqueness of the pulled-back divisor and Galois descent. Mazur rules out the required noncuspidal rational point.

A degree17 point upstairs maps to a point of degree1 or17 on Y. The first alternative already gives a rational point on X_0(103); the second is the irreducible divisor just considered. This handles the exceptional cusp disk without any gonality assumption on Y.

8. THE OTHER LOCAL EXCEPTIONS

For any h fixing the nonrational reduction and C the rational cusp lift, finite image of h-1 implies y=[h(D-C)-(D-C)] is torsion reducing to0, hence2y=0. If y=0 and gon>17+c, or simply gon>2*(17+c), then h(D-C)=D-C as divisors. Disjoint cuspidal and noncuspidal support gives hD=D. The possible noncuspidal orbit lengths are ord(h), ord(h)/2 at j=1728, or ord(h)/3 at j=0. Irreducibility of D rules out any of these lengths except1 or17.

At89 the nonrational cusp has degree11, with complement6. The odd-degree Hecke-isogeny certificate on the h=diamond2 anti-Jacobian, proved on both Betti signs in degree16, gives y=0. Indeed N_h*y=0 and 2y=0 imply y=(11-N_h)*y in (h-1)J, where the odd-degree annihilator is injective on2-torsion. The gonality bound exceeds23, and h has order11, so invariance is impossible.

At97 the supersingular degree16 divisor is fixed by the order4 subgroup. The doubled bound 2*(17+1)=36 is below the gonality lower bound. Thus D is invariant; the possible orbit lengths4 and2 do not divide17.

At109 the supersingular degrees are9 and12, fixed by diamond23 of order18. The doubled bounds50 and44 are below the gonality bound. The possible orbit lengths18,9,6 are incompatible with17.

At113, the degree7 supersingular point, two such points, and the degree14 nonrational cusp are fixed by diamond2 of order14. The doubled bounds 54 and40 are below the gonality bound. The exceptional ordinary degree8 orbit (and its pair exceptions) is fixed by diamond40 of order8; the largest doubled bound is52. None of these orbit lengths is1 or17. In the mixed degree7+8 case diamond98 is an involution fixing the whole reduction. The doubled bound38 is sufficient. Invariance forces j=1728, excluded through degree17 by cm_small.m using Frobenius at3.

At127 the nonrational cusp has degree7 and all supersingular points have degree14. These are fixed by diamond2 of order7. The doubled bound is at most54, below325*(127^2-1)/65536. The ordinary degree9 orbit of size7 is fixed by the order9 subgroup, and its doubled bound

is50. Invariance excludes these cases. The remaining ordinary and
mixed cases, apart from the following partially relative case, are the
retained principal-divisor obstructions.

The mixed degree9 ordinary orbit plus degree7 nonrational cusp and one
rational cusp F is treated using two quotients. Put H9=<diamond(3^7)>
and H7=<diamond2>, of orders9 and7. The ordinary divisor is a pullback
of a rational point on X/H9, and the nonrational cusp divisor is a
pullback of a rational point on X/H7. Both actions are free on the
corresponding geometric support. Use
 alpha=(diamond(3^7)-1)*(diamond2-1)
composed with torsion annihilators at2 and3. At2, the needed hypothesis
holds because rational torsion on J_1(127) is cuspidal, as recorded in
Derickx--Stoll Remark4.4. These are actual integral maps killing J(Q)
and both pulled-back Jacobians. They vanish on
             L=F+pi9^*(X/H9)+pi7^*(X/H7),
a smooth local family of dimension2. There is one relative position of
the two closed orbits under Delta, since their orbit sizes7 and9 are
coprime. All63 positions of F have transverse rank15. The integral
relative lemma forces D into L, contradicting the fixed cusp F.
verify_relative127.py independently reconstructs every cotangent entry,
including the nonrational-cusp coordinates obtained from the nodal
Tate curve with parameter a primitive127th root in F_(2^7).

At97 the operator sigma^12-1 is intentional. The inherited expression
sigma^6-sigma^4-sigma^2+1 does not vanish at a primitive12th root and
cannot supply the asserted finite image. The degree17 computation uses
sigma^12-1 on356 ordinary orbits and the valid operator
(sigma-1)*Phi12(sigma) on the three remaining degree10/14 closed
orbits of size24. The combined passing certificate is also a replacement for the affected
ordinary97 computations in the earlier proof packets; see the separate
level97 correction note.

9. COVERAGE AND REFERENCES

verify_coverage.py verifies that every prime up to the dyadic cap has
a verified occurrence or a complete exclusion, and that every possible
local support multiset is accounted for. verify_ranks.py independently
checks the stored obstruction matrices in Sage. verify_auxiliary.py
checks the dyadic matrices, cusp relation spaces, relative matrices and
finite-image inputs. The stabilizer and CM scripts check the exceptional
automorphism and Frobenius claims. final_coverage.json
records the exact counts and any unresolved cases.

Primary references:
 Derickx--Stoll, arXiv:2505.14109v1,
 https://www.mathe2.uni-bayreuth.de/stoll/papers/asymptotics.pdf
 Siksek, Chabauty for symmetric powers of curves,
 Algebra & Number Theory3(2009),209--236,
 https://msp.org/ant/2009/3-2/ant-v3-n2-p03-p.pdf
The relative argument used here is the integral formal-scheme lemma
proved in Section5, so no odd-residue-characteristic restriction from
an integration-based formulation is being imported.

FINAL VERIFICATION

The complete ledger accounts for all6595 primes through66049,
all223 local support types, and7200 ordinary source orbits.
There are9413 independently verified principal-divisor obstruction

matrices. All remaining rank defects are resolved by the explicit stabilizer or relative arguments above. There are no unresolved primes.